Turtle Graphics

CSCI 1101

2024-02-09

Graphics

Computing with numbers and strings is great and all,
but sometimes we want the computer to show us pictures.

Python includes an easy-to-use drawing library called `turtle`.

In order to use it we need to include this line in our program
or at the interactive interpreter:

```
import turtle
```

This tells python to read in contents of the `turtle` **module**.

Moving the Turtle

By default, the turtle carries a pen in its mouth that draws on the screen.

You can make the turtle:

move forward with

```
turtle.forward(<distance_in_pixels>)
```

move backward with

```
turtle.backward(<distance_in_pixels>)
```

turn left with

```
turtle.left(<angle_in_degrees>)
```

turn right with

```
turtle.right(<angle_in_degrees>)
```

Erasing Your Drawing

You can undo the last thing you drew with
`turtle.undo()`

And erase the whole drawing with
`turtle.clear()`

If you also want to return the turtle to its original state and start over, use
`turtle.reset()`

Screen Coordinates

The turtle's screen has a coordinate system with the *origin* at the center, the x-axis pointing right, and the y-axis pointing up.

You can make the turtle go to a particular point on the screen with

```
turtle.goto(<x-coordinate> , <y-coordinate>)
```

This is useful if you don't want to calculate the heading and distance between where the turtle is and where you want it to go.

Similarly, you can make the turtle face in a particular direction with

```
turtle.setheading(<angle_from_x_axis>)
```

Controlling the Pen

If you want the turtle to move somewhere without drawing a line you can call
`turtle.penup()`

Then when you're ready to do more drawing you can call
`turtle.pendown()`

The default pen draws a 1-pixel wide black line.

You can change this with
`turtle.pensize(<line_width>)` and `turtle.pencolor(<color>)`.

Drawing Shapes

By “shape”, I mean a “closed curve”, where we end drawing at the same place where we start.

Consider the following function:

```
def draw_square(size) :
    turtle.forward(size)
    turtle.left(90)
    turtle.forward(size)
    turtle.left(90)
    turtle.forward(size)
    turtle.left(90)
    turtle.forward(size)
    turtle.left(90)
```

We can make the turtle draw a square with

```
draw_square(100)
```

Activity: Drawing Shapes

Define a function `draw_rectangle` that draws a rectangle with a specified width and height.

Filling Shapes

We can not only draw shapes, but fill them as well.

Set the fill color with

```
turtle.fillcolor(<color>)
```

To fill a shape, just before you start drawing it call

```
turtle.begin_fill()
```

then just after you finish drawing it call

```
turtle.end_fill()
```

Circles

There is a function for drawing a (kind of) smooth circle:

```
turtle.circle(<radius>)
```

```
turtle.fillcolor("red")
turtle.begin_fill()
turtle.circle(100)
turtle.end_fill()
```

Changing Speed

If we want to see more clearly what the turtle is doing, or we just want to hurry up and get to the finished picture we can change the turtle's speed with

```
turtle.speed(<speed>)
```

The speed should be an integer between 0 and 10 with:

1 slowest

2 slightly faster

⋮

10 fastest

0 instant

To Do

Draw a picture with turtle.

Start working on *Project 1 - The Turtle Module* on CodeRunner.
It is due in two weeks.

Finish *Homework 3: Conditionals* on CodeRunner by 11:00AM next Thursday.